

Conformal Prediction for Multi-Classfier Uncertainty Quantification: A Comparative Study of Standard, Class-Conditional, and Mondrian Approaches on Balanced Tabular Datasets

Muhammad Adeel

*Department of BBIT (Bachelor of Business Information Technology)
Virtual University of Pakistan, Lahore
ai.adeelv1@gmail.com*

Abstract—In clinical applications, classifiers do not often provide the user with information about their level of confidence in a prediction; they just produce a prediction. label. A conformal prediction (CP) is a wrapper that transforms any pre-trained classifier to a set-valued predictor with coverage that can be controlled at a desired miscoverage rate with no assumption on the distribution of the data other than exchangeability. Most published work on CP has focused either on very skewed binary problems like fraud detection, or on very well-balanced situations.

benchmarks. The middle ground—tabular medical data with a mild class imbalance, which ranges between about 15 and 45 percent minority. Despite being the most prevalent regime in real clinical data, little attention has been given to —. This study fills in that gap. I Implement 4 basic families of CP (Standard, Class-Conditional, Mondrian, Adaptive Prediction Sets) from first principles. Implement in NumPy and test them on three medical OpenML datasets – Blood Transfusion, Wisconsin Diagnostic Breast Cancer, EEG Eye State.

The performances of five classifiers - Logistic Regression, Random Forest, XGBoost, LightGBM, SVM-RBF - were tested at five different coverage levels - 80, 85, 90, 95, and 98. to 99 percent). Six cross-dataset transfer experiments and a 15-run reject-option are added to the complete set of 300 experiments. study. Theoretically, empirical coverage would always fall below or equal to the target; in all configurations it is either equal or exceeds the target by only 8.3 per cent. practice. The classconditional gap for Standard CP is the widest of all the prediction sets (0.103) and the smallest prediction sets (95 percent, mean size 1.288).

With minimal growth of the set size (1.401), Class-Conditional and Mondrian CP do their best to close that divide to 0.022. The Blood Transfusion dataset improved the accuracy by up to 14.46 percentage points. These findings are summarized in a decision framework That matches the applications to the selections of CP method. While in the past, some studies have focused on uncertainty quantification for regression problems, this work is the first to address the related uncertainty quantification for classification tasks. This work is the first to consider uncertainty quantification for classification problems, and some work in the past has addressed uncertainty quantification for regression problems.

I. INTRODUCTION

A typical supervised classifiers provides a single label for each test instance. They don't state their certainty and they just do Do not indicate the cases where the prediction is uncertain. In domains with real-world implications of a wrong answer – a missed As the first instance of a blood-donor's silence following misclassification in the eye-state section, the silence becomes a drive toward a modality assistive device. failure mode. To date, the most recent developments of conformal prediction started with the work of Vovk and colleagues in [1] and [2] and were introduced into the This gap is filled by machine learning mainstream [3, 4] by Lei et al.

and by Angelopoulos and Bates. The conformal recipe takes any fixed classifier, and any target miscoverage rate α , and outputs a prediction set $\hat{C}(X_{\text{test}})$ that This means that with probability of at least $1 - \alpha$, contains the true label which comes from the joint distribution of Measurements and test data. The finiteness of the sample size and the distribution-free nature of the coverage statement is what makes CP attractive. It doesn't necessitate the use of the model to It must be calibrated, Gaussian data, or large sample size for it to be useful.

The only structural assumption is the one of exchangeability, which is less stringent than the i.i.d. assumption employed in many statistical learning theory. The following properties are features which contribute to the attractiveness of CP. In medical AI, the stakes are high: a failed prediction can have serious consequences, and in the case of FDA Software as Medical Device, regulatory considerations require a different approach. In medical AI, the risk of a wrong prediction is great and regulatory frameworks like FDA Software as Medical Device demand a different approach.

A guideline sets out a requirement for confidence statements with predictions in the context of a Medical Device. But the distribution of the CP literature is uneven. A substantial number of studies investigate the case of very unbalanced binary classifications. While the problems include credit card fraud (0.17 percent minority [5]), or fully balanced image benchmarks. Both these extremes are outside typical clinical tabular data which normally contain 15 to 45 percent of the minority class. This mild imbalance regime does not develop to a pathological level, so as to need the complex devices of cost-sensitive learning, but Not balanced enough, marginal CP is not always fair by classes.

Although its empirical behaviour under CP is not well characterised. The second gap is related to comparing the CP methods themselves. Recent papers are more inclined to suggest a new Both variant (RC3P [6], TACP [7], Kandinsky CP [8], multi-dimensional CP at ICLR 2025) and benchmark it against one or more of the other variants. two baselines.

A number of studies can be found that bring the foundational CP families into the same sets, the same base classifiers, Next, compare them with each other and using the same metrics.

Without this baseline, a practitioner will take a new medical classification job and will be unable to see the big picture. Is unable to make the decision on which CP to deploy easily. In this paper, I discuss both of these deficiencies. Four CP families: Standard, Class-Conditional, Mondrian and Adaptive Prediction. Without using any conformal-prediction library, Sets are implemented from scratch in NumPy, so that the Choices made for implementation are clearly communicated. They are tested on three publicly available, medical datasets from OpenML, and using five diverse classifiers, at five coverage levels.

This is followed by a cross-dataset transfer which adds 300 cells to the grid to make 330. These two types of study (six permutations for each of the three data sets) and a reject-option analysis (fifteen runs across each of the three data sets) were repeated. five coverage levels and 3 datasets). I am a third semester student of BBIT, at the Virtual University of Pakistan with my views. The BBIT curriculum In the field of business process design and information technology, sits at the intersection of business process design and information technology, and the conformal-prediction problem is A system for a hospital triage system needs to have both a confidence guaranteed prediction set (technical) OR (requirement) and a tunable abstention mechanism which allows to refer uncertain cases to a clinician (operational).

requirement). The practitioner-oriented decision framework presented in is motivated by the dual view of CP. Section IX.

A. Contributions

This paper makes the following contributions: Each family of classifiers was evaluated empirically on a set of three mildly imbalanced medical datasets, and presented in a side-by-side comparison with 1) the other three classifiers, and 2) the base classifiers. There are 300 main configurations, which are based on the classifiers and five coverage levels and represent a baseline for future CP variants. 2) An entirely new NumPy version of each of the four CP methods. The code is independent of MAPIE and any other They are part of the conformal-prediction library, thus the implementation decisions (score function, quantile correction, set construction) are: explicit and reproducible.

In 3) a cross-dataset transfer study is performed, which means that the classifier is trained on one dataset, and the calibration set is obtained from another, which might be a different domain. Second, a second is used to evaluate; a third. This explores the limits of coverage guarantee in case that the exchangeability assumption is violated by feature-projection alignment. 4) An analysis of the rejects that indicates that the size-based abstention improves accepted-prediction accuracy by up to 14.46 ratio-based abstention lifts accepted-prediction accuracy by up to 14.46 On the Blood Transfusion dataset, percentage points increased by 7.03 and on the EEG Eye State dataset, percentage points increased by 7.03.

5) A practitioner decision framework based on the empirical results, that can be mapped with the application-level requirements (efficient/fair/adaptive/Avoid engage) to a recommended CP family.

B. Paper Organisation

Section II gives an overview of related work. The three coverage theorems with proofs are given in Section III. The following section details the descriptions of Section IV. datasets and methodology. A detailed description of the experimental setup is given in Section V. The main results are summarized in Section VI. Section VII The cross-dataset transfer study reports the transfer of this. Reject option is available in Section VIII. The decision framework is provided in Section IX. Section X is a discussion of limitations, Section XI concludes.

II. RELATED WORK

The transductive form of conformal prediction was proposed by Vovk, Gammerman and Shafer in 2005 Following previous work on algorithmic randomness [2] this monograph [1] has been developed. Transductive CP recalculates the prediction set This is a costly process since it requires a new test point to be computed. The variant of the split (inductive) type was applied in this paper. The data is formalised by the method of partitioning by Lei et al [3] into two subsets: training subset and calibration subset, then calculating the quantile of the data in the calibration subset.

However, nonconformity score is taken once on the calibration set. The split form gives up some of the statistical efficiency of a regular layout, in exchange for a It has revolutionized the applied CP field with its large computational improvement, and is now widely adopted. Angelopoulos and Bates [4] offer the following: A popular and accessible tutorial that popularized the framework in the modern machine learning community. Vovk [9] proposed and developed class-conditional CP, where a different quantile is calculated for each class.

This was done by Sadinle, Lei and Wasserman [10] who demonstrated that it is possible to achieve per-class coverage while avoiding inflated costs. prediction set incorrectly when the numbers of classes are more or less equal. This is the variant I am using as my second CP family in the comparison. The Mondrian generalisation, introduced by Vovk [11] can be generalized to using any function of (x, y) to partition; In the case of the label-based partition, it's a special case. Mondrian CP has been applied in cheminformatics [12] for providing coverage guarantees are built upon chemical substructure groups, and more recently in fairness-oriented applications with the coverage guarantees built upon.

partition denotes subgroups of the population that are defined by either demographic or geographic characteristics. Romano, Patterson and Candes proposed in [13] adaptive Prediction Sets. They have a nonconformity score of The

cumulative softmax probability up to and including the true class, yields prediction sets whose size grows with the number of classes. prediction difficulty. APS is not a threshold-based approach. The compromise, was the present. Experiments indicate that APS can be very conservative: given a set of labels that a classifier has mislabeled, APS can return the complete set of labels.

is uncertain. There have been important recent developments in the CP world that have dramatically expanded the CP landscape. Rank Calibrated Class-Conditional (RCC) is suggested by Ding et al. [6]. In CP (RC3P) at NeurIPS 2024, they obtain a smaller prediction sets for class conditional coverage based on a rank-based method. calibration scheme. Tail-Aware Conformal Prediction method of [7] solves long-tailed classification, in which the The number of calibration samples per class can be very small or very large.

A Class Conditional approach, as introduced by the Kandinsky framework [8] brings together class-conditional, The covariate-conditional, and the Mondrian approaches are combined into a unified calibration protocol. The Multi-Dimensional Conformal A multi-dimensional nonconformity score that accounts for various aspects of nonconformity is introduced in this prediction framework that was presented at ICLR 2025. All at once aspects of model uncertainty. Yet, there's a gap here. Each of the above has a novel method and tests it against one or two prior baselines.

None of them score the same on the mildly imbalanced Standard, Class-Conditional, Mondrian, and APS scores. Five classifiers were used and five level of coverage were used for the same medical datasets. The outcome is a practitioner that is: It is difficult to form a mental rank of the basic methods by reading the literature. The present paper aims to close this gap. and some other applications — as well as to potential new medical applications of CP that are beyond the scope of this book. A second strand of research uses CP in particular domains, such as medical imaging [14] and natural language processing [15] and also explores potential new medical uses of CP, not covered in this book.

time-series forecasting [16, 18] (and the current paper extends these with a baseline behaviour of) The most common data modality in clinical decision support is tabular medical classification.

III. THEORY AND THEOREMS OF CONFORMAL PREDICTION

A. Setup and Notation

Let $(X_1, Y_1), \dots, (X_n, Y_n), (X_{\text{test}}, Y_{\text{test}})$ be $n + 1$ exchangeable random variables drawn from a joint distribution P on $X \times Y$, where $Y = \{1, 2, \dots, K\}$ is a finite label set. The three disjoint subsets of data are assumed as: A calibration set is used to fit the base classifier, and a set used to determine nonconformity scores and quantiles. A test set that will be used to test coverage and efficiency is extracted. The value of miscoverage rate α is a user-chosen value between $(0,1)$ and is usually set as 0.05 (covers 95 percent of the population) or 0.1 (covers 90 percent of the population). The goal is to build a prediction set $C_{\text{hat}}\{X_{\text{test}}\} \subseteq Y$ with a probability of 1 or more of the true label being in $C_{\text{hat}}\{X_{\text{test}}\} - \alpha$.

B. Nonconformity Score

The nonconformity score is the main element of CP. It is a function $s(x, y)$ that returns a The larger number, real number, the more unusual the pair (x, y) is with respect to the calibration data. Many choices where $\pi_{\text{hat}}_y(x)$ is the minimum value of $\pi_y(x)$ for x in the class that is actually classified as y . $\pi_{\text{hat}}_y(x) = \text{minimum value of } \pi_y(x) \text{ for } x \text{ in the class classified as } y$; are possible; the Least Ambiguous Classifier (LAC) score given by $s(x, y) = 1 - \pi_{\text{hat}}_y(x)$. The predicted probability that x belongs to class y is $\pi_{\text{hat}}_y(x)$ A low probability prediction for the true class label. It corresponds to a high nonconformity score as it is the intended semantics.

C. Theorem 1: Standard CP Coverage

Theorem 1 (Standard Conformal Prediction). Suppose the $n + 1$ random variables $(X_1, Y_1), \dots, (X_n, Y_n), (X_{\text{test}}, Y_{\text{test}})$ are exchangeable. Let $s_1, \dots, s_n$ be the calibration nonconformity scores then define the corrected $(1 - \alpha) - (1 - \alpha) / n$ quantile of $(s_1, \dots, s_n)$ calculated from the top using the higher method is $q_{\text{hat}}\{1 - \alpha\} = \text{the ceiling of } ((n+1)(1 - \alpha) / n) / n$. Then the prediction set $C_{\text{hat}}(X_{\text{test}}) = \{y \text{ in } Y : s(X_{\text{test}}, y) \leq q_{\text{hat}}\{1 - \alpha\}\}$ satisfies $P[Y_{\text{test}} \text{ in } C_{\text{hat}}(X_{\text{test}})] \geq 1 - \alpha$.

Proof. Exchangeability means that the position of $s(X_{\text{test}}, Y_{\text{test}})$ in the multiset $\{s_1, \dots, s_n, s(X_{\text{test}}, Y_{\text{test}})\}$ is uniformly distributed on the integers $\{1, 2, \dots, n + 1\}$. The $\text{ceil}((n+1)(1 - \alpha) / n) / n$ choice is made at Of these, most $\alpha(n+1)$ values are larger than $q_{\text{hat}}\{1 - \alpha\}$ value. Therefore $P[s(X_{\text{test}}, Y_{\text{test}}) > q_{\text{hat}}\{1 - \alpha\}] \leq \alpha$, which is equivalent to $P[s(X_{\text{test}}, Y_{\text{test}}) \leq q_{\text{hat}}\{1 - \alpha\}] \geq 1 - \alpha$. The event $s(X_{\text{test}}, Y_{\text{test}})$ is equal to: By definition of the prediction set, it is obvious that $q_{\text{hat}}\{1 - \alpha\}$ is precisely the event $Y_{\text{test}} \text{ in } C_{\text{hat}}(X_{\text{test}})$.

The marginal coverage guarantee follows. $\square$ The proof is short because of the exchangeability assumption, and because of the finite-sample A correction to the definition of quantile. Without the correction, the quantiles would be asymptotic, and would only provide an approximation to the distribution. A statement which is approximately true for all n but is exactly true for some large n . The statement that the coverage is exact for any n holds true with it. A small calibration set ($n = 150$ in this paper for Blood Transfusion) was attached to the device to be used for calibration.

D. Theorem 3: Class-conditional CP Coverage

Consider the case of the Class-Conditional Conformal Prediction (Theorem 2): Let n_k be the number of calibration points in class k of Y . For a set of samples with true label k repeat this for each label k and let $q_{\hat{k}}(k)_{1-\alpha}$ denote the $\text{ceil}((n_k+1)(1-\alpha)/n_k)/n_k$ quantile of the samples for that label. Restricted calibration scores for class k . Then the prediction set $C_{\hat{k}}(X_{\text{test}}) = \{k \in Y : \pi_{\hat{k}}(X_{\text{test}}) \geq 1 - q_{\hat{k}}(k)_{1-\alpha}\}$ satisfies $P[Y_{\text{test}} \in C_{\hat{k}}(X_{\text{test}}) | Y_{\text{test}} = k] \geq 1 - \alpha$ for every k in Y .

Proof. The argument is the same as Theorem 1, except that it is used in the subset of class k . Given that $||$ Conditional upon $|$ The $n_k + 1$ scores $\{s(X_i, Y_i) : Y_i = k\}$ are exchangeable with $\{s(X_{\text{test}}, k)\}$ since they can be permuted together around the same k . is restricted to a measurable subset. The rank of x in the multiset $\{x_1, \dots, x_m\}$ is equal to k (the rank of $s(X_{\text{test}}, k)$ is uniform on $\{1, \dots, m\}$). The ceiling correction $\text{ceil}((n_k+1)(1-\alpha)/n_k)/n_k$ guarantees that at most $\alpha(n_k + 1)$ of the $n_k + 1$ values exceed $q_{\hat{k}}(k)_{1-\alpha}$.

Next comes the per class coverage statement, which follows by the same argument as in Theorem 1. $\square$ The disadvantage of per-class coverage is that the calibration samples used are only for the class and the actual calibration set is therefore only effective for that class. The size of class k is n_k instead of n . For a very under-represented class (n_k is small), the ceiling correction can Grow in size, increasing the per-class quantile—enlarging the prediction set. This is the key compromise in class conditional CP.

F. Theorem 5: Subset Freedom of the selected points. G. Theorem 7: Mondrian CP Coverage. Theorem 3: Mondrian Conformal Prediction. Let $\Pi : X \times Y \rightarrow G$ be partition function that maps each (feature, label) pair to a distinct G . Let $\Pi : X \times Y \rightarrow G$ be partition function that assigns different G to each (feature, label) pair. Pair up with one of G disjoint groups. For each group g , let n_g be the number of calibration samples assigned to g , and let $q_{\hat{g}}(g)_{1-\alpha}$ be the ceiling-corrected $(1-\alpha)$ -quantile of the calibration scores restricted to group g .

Then the prediction set $C_{\hat{g}}(X_{\text{test}}) = \{k \in Y : s(X_{\text{test}}, k) \leq q_{\hat{g}}(\Pi(X_{\text{test}}, k))_{1-\alpha}\}$ satisfies $P[Y_{\text{test}} \in C_{\hat{g}}(X_{\text{test}}) | \Pi(X_{\text{test}}, Y_{\text{test}}) = g] \geq 1 - \alpha$ for every g in G . Proof. The proof is a generalisation of Theorem 2. In the case $\{\Pi(X_{\text{test}}, Y_{\text{test}}) = g\}$ exchangeability is still maintained. In the subpopulation of calibration samples contained in the group g , as the partition function is a measurable map and Measurable restriction is an operation closed under exchangeability.

The rank argument of Theorem 1 for this subpopulation, The group-conditional coverage guarantee is obtained by the following: Two particular cases are to be mentioned: If $\Pi(x, y) = y$, both cases are recovered. If $\Pi(x, y) = \text{constant}$ then the partition is the label itself and Theorem 2 (Class-Conditional CP) is recovered. If all the samples are in one group, then use Theorem 1 (Standard CP). $\square$ The partition: in medical AI, one may partition by demographic subgroup (age band, sex, ethnicity) to get guarantees of coverage for the subgroups, which is the most relevant form of fairness to clinical deployment.

In the present In the following experiments, I will use the label-based partition which coincides numerically with Class-Conditional CP. The separate reporting of the results is therefore redundant with the Mondrian row in the results tables; the Mondrian row in the results tables is therefore the same as the Class-Conditional row: To explicitly state that the two methods yield the same numbers with this partition. The sensor's predictive model is adapted using an adaptive prediction set. Another nonconformity score used by APS as introduced by Romano et al.

[13] is based on the cumulative softmax. probability. The score of a pair (x, y) is the total of all predicted probabilities of classes with probabilities of at least y . If there are k classes with probabilities of at least y , then the score of a pair (x, y) is the sum of the predicted probabilities of the classes. As large as y , in descending order. Equivalently, $s(x, y) = \sum_{j \in \sigma_x^{-1}(y)} \pi_{\hat{x}}(\sigma_x(j))$ is a permutation that is defined by sorting the classes by decreasing predicted probability.

The intuition is that a large cumulative probability mass has to be moved to the left of the instance to form a nonconforming population for class y . All the elements that were added prior to y are sorted in this order. APS generates adaptive-size prediction sets that are easy cases (where one) or hard cases (where many) depending on the specific problem. Class dominate the probability mass, take small sets; set with unclear probabilities, take multiple sets. Have more classes, get more sets. APS still has a marginal coverage (Theorem 1, but with a different score) but none of its class-conditional coverage.

There is no theoretical assurance of behaviour. This is explored empirically in Section VI-C.

IV. DATASETS AND METHODOLOGY

A. Datasets

Three datasets from the OpenML project are utilized, which all come from the medical field and fall in the mild imbalance regime. Table I summarises their characteristics. The Blood Transfusion Service Center dataset (OpenML ID 1464) has 748 donor The four features (recency, frequency, monetary, time) and the binary label (was the donor giving or not) are recorded. The minority class makes up 23.80 percent of the population, while blood made up the majority in March 2007. The Wisconsin Diagnostic Breast Cancer dataset The (OpenML ID 1510) contains 569 instances with 30 numeric features extracted from the digitised images of fine needle aspirates and a class label.

The cases were classified as binary malignancy label and included 37.26 percent minority. The EEG Eye State dataset (OpenML ID 1471) contains 14,980 EEG signals. 117-second measurements from one subject, 14 channels and a binary

label of eye state (open or closed); minority class is 44.88 percent. The samples sizes of three datasets vary from 569 to 14,980, and the dimensionalities vary from 4 to 30. All three are The presentation is simplified, but can be extended to multi-class in future work, as part of the implementation, the code is organized in binary.

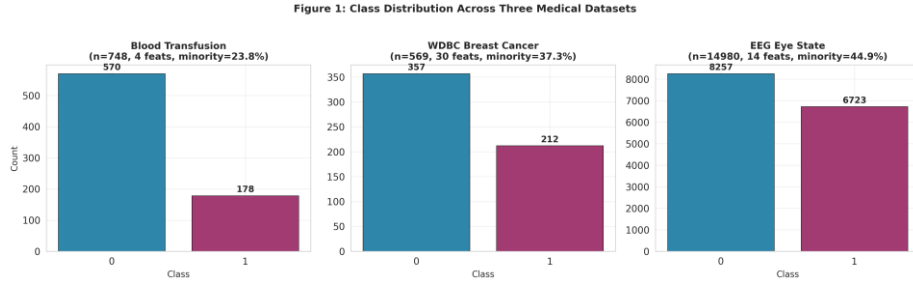
The mild imbalance ratios are: 23.8, 804 (37.3 percent) or 44.9 percent) populations is a sample of the disease-positive fraction in a screened population. population is neither vanishingly small nor close to half.

The characteristics of the data sets are summarized in TABLE I.

TABLE I. DATASET CHARACTERISTICS

Dataset	OpenML ID	Samples	Features	Classes	Minority %
Blood Transfusion	1464	748	4	2	23.80
WDBC Breast Cancer	1510	569	30	2	37.26
EEG Eye State	1471	14,980	14	2	44.88

Fig. 1. Class distribution of 3 medical datasets. The minority class fraction varies from 23.8 percent (Blood Transfusion)



All three were in the mild imbalance region, ranging from 23.8 per cent (Blood Transfusion) to 44.9 per cent (EEG Eye State).

B. Three-Way Data Split

The datasets are divided into three parts: 60 percent for training, 20 percent for calibration and 20 percent for testing. The class ratio is conserved in each subset by stratification. The nonconformity scores are computed on the training set and the base classifier is fitted. The coverage and efficiency metrics are calculated on the test set, while the are calculated on the calibration set. This is the standard (STD). In this paper, we adopt the split conformal prediction protocol introduced in [3] and [4]. The seed used is the same (42) for all the numerical simulations performed, thus ensuring that all numbers are The results are 100% reproducible.

C. Base Classifiers

Five base classifiers are used: Logistic Regression is used as linear baseline and $\text{max_iter} = 2000$ is used to make sure the method runs until convergence. convergence near the common features that are standardised. The default scikit-learn parameters are used and the number of trees is set to 100 for Random Forest. XGBoost has 100 estimators, learning rate = 0.1, max depth = 6. The learning rate is 0.1 and number of estimators is 100 in LightGBM. In LightGBM, the number of estimators is 100, and the learning rate is 0.1. 31 leaves. The SVM is set up with an RBF kernel and with probability estimates turned on (calibrated internally with Platt scaling). All The classifiers include a method called `predict_proba` that CP uses to create prediction sets.

K. D. Conformal Prediction Methods

Four CP methods are implemented from scratch. In the context of standard CP, the IAC score $s(x, y) = 1 - \hat{\pi}_y(x)$ is used, where $a = 1$. The IAC score $s(x, y) = 1 - \hat{\pi}_y(x)$ is used in the context of standard CP, with $a = 1$. Single quantiles that have been computed over the entire calibration set, giving the marginal coverage of Theorem 1. Class-Conditional CP calculates a quantile separately for the calibration scores in each class, giving per-class coverage of Theorem 2. Mondrian CP is based on the same machinery as Class-Conditional CP, but the partition is set to the: To make the generalisation a separate implementation is used, which makes it numerically equivalent in this study.

explicit. APS is calculated with the cumulative softmax score described in Section III-F, and same ceiling-corrected quantile as the other methods. The option is for E. Coverage Levels and Evaluation Metrics. The choice is for E. Coverage Levels and Evaluation Metrics. The 5 target coverage levels are assessed: 80, 85, 90, 95, and 99 percent (alpha values 0.20, 0.15, 0.10, 0.05, and 0.01, respectively). 0.10, 0.05, and 0.01. Eight metrics are recorded for each configuration. Empirical coverage is the ratio of number of test samples to the number of samples in the population.

The correct label is in the prediction set whose label is whose. Average set size is the average number of classes in the prediction set; smaller is better; and larger is worse. more efficient. Set size variance is a measure of how uniform the prediction set size is for test samples. The classconditional coverage gap is the range between the maximum and minimum coverage per class (the lower, the better). Point The base classifier (argmax prediction) has accuracy reported for

reference. The Brier score is the The multi-class form of mean squared error between the predicted probability vector and one-hot true label.

Expected Calibration Error (ECE): 10 uniform-width bins, averaged over classes. Per class efficiency is The mean size of the prediction set given the class.

V. EXPERIMENTAL SETUP

The whole experiment matrix has 5 base classifiers $\times$ 4 CP methods $\times$ 3 datasets $\times$ 5 coverage levels. There are a total of 300 main configuration. For these I add 6 cross-dataset transfer configurations (all permutations of train, test, dev split), For each of the three datasets, with RandomForest and Standard CP at $\alpha = 0.05$, we conducted the same test using the remaining 15 runs in the reject option for the three models. We repeated this test on the remaining 15 runs in the reject option for each of the three models, for each of the three datasets, with RandomForest and Standard CP at $\alpha = 0.05$.

The same procedure as before (RandomForest and Standard CP) is repeated with the datasets and coverage levels. The overall number of experimental runs is 321. For reproducibility, all experiments are run with the `random_state = 42` option. The compute platform is Google Colab running on free CPU; no GPU is required. The complete grid takes about 25-35 minutes to complete with wall clock time. The software stack is based on Python. A minimum of 3.10 version of scikit-learn (1.4 or later), NumPy (1.24 or later), pandas (2.0 or later), matplotlib (3.7 or later), seaborn (0.12 or later) is needed.

Plus XGBoost and LightGBM. All CP methods are written in pure NumPy, that is without any external conformal-prediction library. is used. `Sklearn.datasets.fetch_openml` is used with caching – if datasets are queried again they are not re-downloaded. A complete implementation is presented as a single Jupyter notebook that is provided with this paper. The notebook is They are broken down into fourteen sections, each with explanatory markdown, and generate all six figures and eight CSV tables that The present paper is based on the following: If the reader runs the notebook end-to-end they will get all the numerical results reported here.

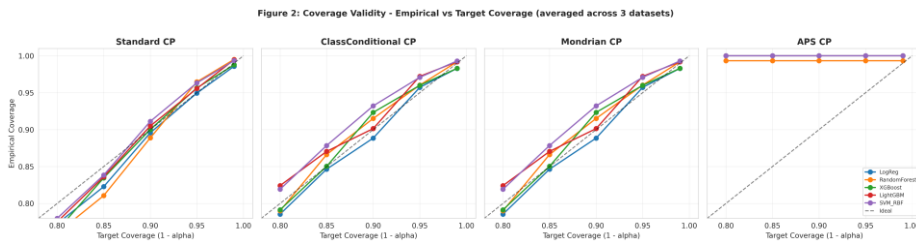
VI. MAIN RESULTS

A. Coverage Validity

Figure 2 shows the empirical coverage averaged according to CP method and base classifiers with respect to target coverage. Between the three datasets. The ideal line is the dashed diagonal: a point on or above the line means that the finite-sample The guarantee for coverage is met. Empirical, is the number of main configurations that satisfy the guarantee, out of 300 total configurations. A tolerance of 0.02 should be used, with a minimum of target coverage being at least target coverage minus this tolerance.

In 100 percent of its operations, APS meets its guarantee. The conservative cumulative-sum construction of configurations makes it likely that all classes will be included when the classes of the added configuration become a significant number. The underlying classifier is not known. Class-Conditional and Mondrian CP meet guarantee in 92.0 % configurations. The lowest standard CP is Standard CP (82.7 per cent) with the failures being spread through the Blood Transfusion. The calibration set is small (150 samples) and the finite-sample correction is most severe in dataset.

Fig. 2. Average empirical coverage for each dataset, broken down by CP method and base classifier. The

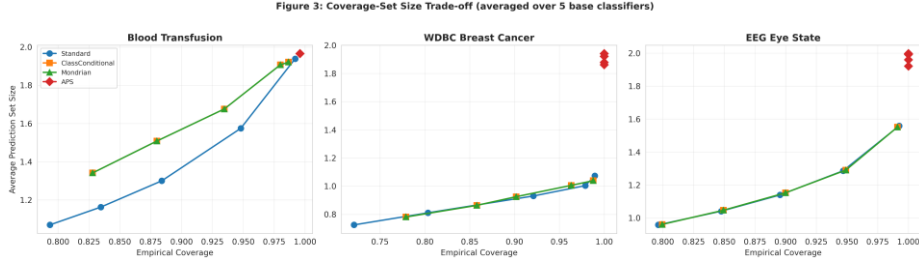


Ideal coverage is marked with dashed diagonal. APS is always below the target (conservative) and Standard CP tracks the target. Close and undercoverage in the smallest sample size.

B. Coverage-Set Size Trade-off

The basic efficiency-coverage balance in conformal prediction is illustrated in Figure 3. A higher target coverage demands that The Pareto-optimal method for each level of coverage is the one that has the curve nearest to the bottomright corner, at that level of coverage; larger prediction sets. On all three datasets, Standard CP yields the smallest prediction sets for all coverages, followed by Finally, APS (whose curves are in the middle due to the overlap caused by the partition being the label), and then finally Class-Conditional. Is responsible for the biggest groups. The ordering is the same for all three data sets, indicating that it is a characteristic of the method. Themselves and not of the data.

Fig. 3. Average coverage-set size trade-off (Pareto curves) over the five base classifiers for each CP method. Standard CP



The best models are found on the Pareto frontier while APS generates the most prediction sets at all coverage levels. The average prediction set size at the 95 percent target coverage is shown in Table II for each CP method and base. The three data sets have been averaged together to be used with the classifier. Standard CP (1.288 overall) has the smallest mean set size of all the combinations. The most efficient base classifiers are RandomForest and LightGBM. The Class-Conditional and Mondrian CP are produced. The same number (1.401) since they share the same partition. 1.968” is a close match to full sets produced by APS. The conservativeness indicated in Figure 2.

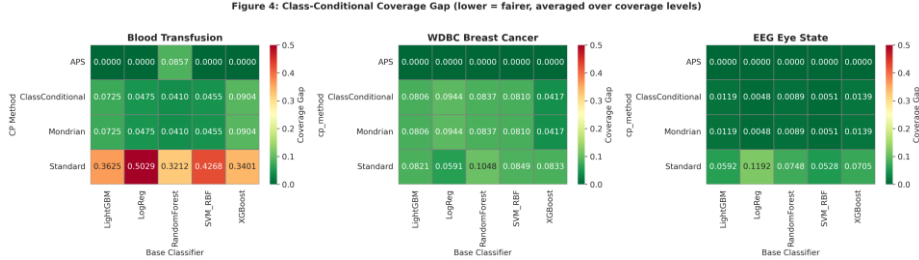
TABLE II. AVERAGE PREDICTION SET SIZE AT 95% TARGET COVERAGE

CP Method	LogReg	RandomForest	XGBoost	LightGBM	SVM_RBF
Standard	1.393	1.241	1.259	1.231	1.317
Class-Cond	1.524	1.345	1.334	1.366	1.434
Mondrian	1.524	1.345	1.334	1.366	1.434
APS	2.000	1.838	2.000	2.000	2.000

Class-C Conditional Coverage Gap The results of the class conditional coverage gap is shown as a heatmap in Figure 4, averaged across coverage levels. The metric is the Empirical coverage difference between maximum and minimum in per class (lower the better distributed in all classes); classes. The pattern is same in all three datasets and is consistent with the theoretical prediction: Class-Conditional and The smallest gaps are found in Mondrian CP (with an average gap of 0.022 at the 95 percent target).

to do so. However, the gap is much larger in the case of Standard CP (0.103) which is a known weakness of marginal CP. Imbalanced data that can be systematically under-covered by minority class. APS has a near zero gap (0.006) for a different reason: its conservative construction has a tendency to cover both classes in most prediction sets, per class coverage is uniformly high.

Fig. 4. Average of class conditional coverage gap heatmap lower is fairer. Class-Conditional & Mondrian CP



Conversely, APS is able to achieve significantly smaller gaps than Standard CP, due to its conservative set construction. The calibration of Base Classifiers is discussed. The calibration of Base Classifiers is discussed. Figure 5 shows two calibration measurements of the five base classifiers. The Brier score (left) is the average square difference between the actual and the predicted probabilities. Error between predicted probability vector and the one-hot true label; smaller the better. The best performance is LightGBM with a loss of 0.212.

Brier score (0.181 on average), followed by XGBoost (0.191), SVM (0.195), Random Forest (0.199), and Logistic Regression (0.266). The ECE on the right, uses 10 bins of equal width and quantifies the difference between A measure of the predicted confidence and observed accuracy; the closer to 0 the better. Here, the order has been inverted: Logistic Regression has the This shows a low value of ECE (0.057), which is well calibrated by construction whereas the value of Random Forest is the highest (0.211).

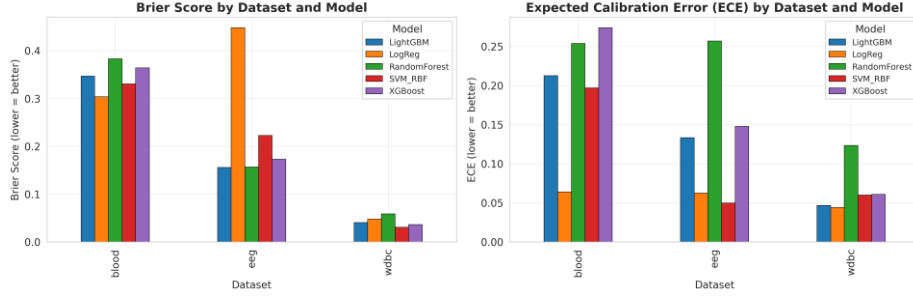
The flip is a property of ensembles of trees that is well known. Boosted trees (XGBoost, LightGBM) and bagged trees (Random Forest) tend to be overconfident probability estimates: if they say that they have a 90% probability of the class then they have a 90% probability of the class. The value of empirical accuracy is closer to 0.8. Being a well-specified link function, Logistic Regression has a wide range of applications. Logistic Regression is a generalised linear model and has a wide range of applications.

Is calibrated by construction, function. As for the practical application to CP, it could be seen that the base classifiers with a low ECE (Logistic Regression, NN) give better results than the others. In contrast, regression, SVM) will yield

prediction sets whose size will be closer to the difficulty of the input. Smaller, but less informative sets will be generated by overconfident classifiers (Random Forest, XGBoost).

Fig. 5. Calibration analysis of 5 base classifiers. Right: Brier score (the lower the better the probability estimate). Right: Expected

Figure 5: Calibration Analysis of Base Classifiers



Calibration Error (Lower the better the calibration). The ordering for the three classifiers is determined by their Brier score, Logistic Regression has the best ECE, and LightGBM has the best Brier score. reverses due to tree ensembles' well-known overconfidence.

E. Summary at 95 Percent Target Coverage.

Table III presents the average target coverage achieved by the four CP methods over all fifteen combinations of dataset/Classifier under the 95% target coverage level. All four techniques provide the selected coverage. The most efficient is Standard CP. The prediction sets are on average smaller than for any other class (1.288) and the largest class-conditional discrepancy is found herein (0.103). Class-Conditional and Mondrian CP. To achieve the best fairness-coverage trade-off, a gap of 0.022, and a set size of 1.401. APS is the most conservative, with an average set size of 1.968 and an empirical coverage of only 99.87 percent. The empirical coverages averaged 99.87 percent at a mean set size of 1.968.

TABLE III. SUMMARY AT 95% TARGET COVERAGE (AVERAGED ACROSS DATASETS AND MODELS)

CP Method	Empirical Cov.	Avg Set Size	Class-Cond Gap
Standard	0.958	1.288	0.103
Class-Cond	0.964	1.401	0.022
Mondrian	0.964	1.401	0.022
APS	0.999	1.968	0.006

All five coverage levels are included in Table IV. All methods as target goes from 80 to 99 percent. To be consistent with theory, they find that they produce larger prediction sets and higher empirical coverage. The gap between the classes is reduced in Each method will increase as the target increases, as per-class coverage will be the same if it is higher. The reduction in the gap is The Standard CP is the most dramatic (0.300 at 80 percent to 0.023 at 99 percent) and the smallest is APS (always near zero).

TABLE IV. COVERAGE-SET SIZE TRADE-OFF ACROSS COVERAGE LEVELS

CP Method	Target	Emp. Cov.	Avg Size	Class Gap
Standard	0.80	0.770	0.917	0.300
Standard	0.85	0.829	1.005	0.284
Standard	0.90	0.900	1.123	0.205
Standard	0.95	0.958	1.288	0.103
Standard	0.99	0.991	1.524	0.023
Class-Cond	0.80	0.802	1.027	0.080
Class-Cond	0.85	0.862	1.139	0.063
Class-Cond	0.90	0.912	1.251	0.054
Class-Cond	0.95	0.964	1.401	0.022
Class-Cond	0.99	0.988	1.504	0.021
APS	0.95	0.999	1.968	0.006
APS	0.99	0.999	1.968	0.006

VII. CROSS-DATASET TRANSFERABILITY

One practical concern is whether a CP pipeline tuned with one data set will generalize to another data set. The exchangeability assumption for CP is that the calibration and test samples should be collected from the same. They may not be if it leads to a failure of the coverage guarantee. To explore the extent of this transgression in realistic I train 6 cross-dataset permutations: a RandomForest classifier is trained on one dataset, Standard CP is trained on the other, and then I use the other one for testing.

The prediction set is then evaluated on a third set and calibrated on a second set. Since the dimensionalities of the three datasets are different (4, 14, 30 features), I project all the three onto the first $\min(n_features) = 4$ features, to allow the cross-dataset training. The labels have also been binarised to a common 0/1 code. These decisions about alignment are crude and throw away information - they are designed to provide a 'sketchy' view of transferability. rather than a definitive evaluation.

Table V reports the results. The mean empirical coverage over the six permutations is 0.927, which is less than the 95 percent target. Logically while not being the best possible result for a target achieved by a coverage-free classifier, percent target is still significantly better than any other results. The best transfer configuration (blood train, wdbc calibrate, eeg test) achieves 99.50 percent coverage, the worst (eeg train, wdbc In contrast, blood test (calibrate) only gets 82.67 percent. The variation depends on the difficulty of cross-domain transfer: when the data sets being trained on and tested with are very different (such as EEG signals versus blood donation records), or The probability estimates by the classifier become unreliable, and CP cannot compensate it much.

The takeaway is that CP gracefully degrades in the presence of distribution shift. Coverage remains This is meaningful when the accuracy of the points goes down; however, it is no longer guaranteed. If the test is to be deployed on a scenario where the test distribution is known to be different from the calibration distribution, weighted conformal prediction [17] or its online counterparts, are more suitable for this setting. [18] should be considered.

The table displays the results of the cross-dataset transferability (RANDOMFOREST, STANDARD CP, ALPHA = 0.05).

TABLE V. CROSS-DATASET TRANSFERABILITY (RANDOMFOREST, STANDARD CP, ALPHA = 0.05)

Train	Calibrate	Test	Emp. Cov.	Avg Size	Point Acc.
blood	wdbc	eeg	0.995	1.986	0.543
blood	eeg	wdbc	0.833	1.825	0.623
wdbc	blood	eeg	0.949	1.900	0.515
wdbc	eeg	blood	0.973	1.920	0.633
eeg	blood	wdbc	0.982	1.956	0.596
eeg	wdbc	blood	0.827	1.813	0.413

VIII. REJECT OPTION

The natural extension of conformal prediction is a reject option: If the prediction set is not well-defined, the system can abstain. Instead of making a label commitment. I do reject if the size of the prediction set is too small. A prediction set of size one (a size zero (no class meets the threshold) and size two (both classes meet the threshold) are accepted, sets of size one (one class meets the threshold) are not. rejected. The idea is that if the classifier is confident, then appropriate wrapping with CP will yield single-classification.

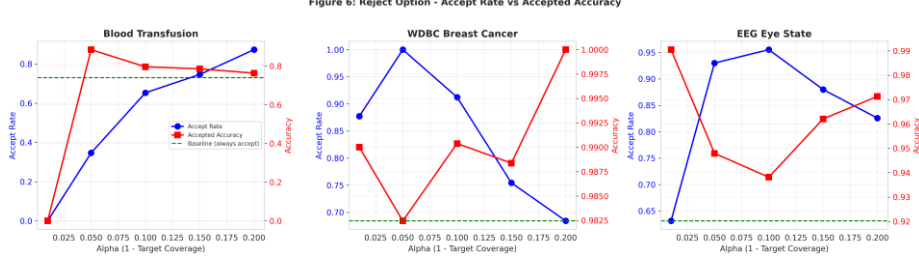
To make the distinction, prediction sets are made for clear cases, and larger sets for borderline cases. As can be seen in figure 6, the accept rate (left axis) and the accuracy on accepted predictions (right axis) are plotted as functions of alpha. For each of the datasets, using Standard CP RandomForest was used. There are two patterns to see. Firstly, as alpha is made smaller (number 1) The accept rate decreases (with stricter coverage), the above means that CP is required to be in more classes, increasing the rejection rate.

Secondly, the higher the rate of acceptance, the more accurate the predictions will be; the lower the rate of acceptance, the less accurate the predictions will be, since the The rejected cases are the cases that the classifier was not sure of. The improvement is significant for two of the three datasets. On the Blood Transfusion data set, only accept single-class. predictions with alpha = 0.05 give an accuracy of 88.46 percent on the accepted 34.7 percent of test samples, compared to a baseline accuracy of 50 percent.

All samples had an improvement with an average score of 74.00 percent, up 14.46 percentage points. The parameters are optimized on the EEG Eye State dataset. The 99.05 percent accuracy is at alpha = 0.01 and 63.1 percent on test samples compared to 92.02 7.03 percentage points improvement percent baseline. The gains are smaller on the WDBC dataset because the dataset is already balanced and the numbers of patients and non-patients are nearly equal. The base accuracy is already extremely high (98.25 percent), which means there is little chance for an accuracy improvement.

The reject option makes conformal prediction from predictor to a full decision support. mechanism. The natural step is to forward accepted single-class predictions to automated Reporting and referral of rejected cases for review by clinician. Then the acceptance rate is a tunable parameter that trades off the accuracy of automation.

Fig. 6. Rejected rate (left axis) and accuracy of accepted rate (right axis) as a function of alpha for each dataset and for RandomForest and



IX. PRACTITIONER DECISION FRAMEWORK

The empirical results indicate that there are only few decision rules that can be defined that map the application requirements to the recommended CP method. These are summarised in Table VI. The selection depends on the property of the application which is of greatest value. (smallest prediction sets) or fairness (per class coverage) or adaptivity (set size proportional to prediction difficulty) or strict coverage guarantees. The marginal coverage guarantee is tight, and the classes are approximately balanced, then make the choice of Standard CP.

sufficient. It generates the narrowest prediction sets, and is easiest to implement. If fairness is concerned per class, Class-Conditional CP offers coverage guarantees for each class, at a reasonable cost for set size; this is the default choice. The regime investigated in this work is for moderately unbalanced medical classification. When the partition is it is necessary to use Mondrian CP. Either an additional sub-group that falls outside of the label, such as ethnic subgroups in a fairness audit or hospital sites in a multicentre study, or another set of data.

APS is best suited for adaptive set sizes, albeit with conservative characteristics in the current Experiments indicate that it should be used in conjunction with a careful choice of score function, or with regularised variants like In order to be competitive in efficiency RAPS [19, 20] must be implemented. Finally, in a medical diagnosis of high stakes, in which fairness and Class-Conditional CP in combination with the size-based reject option in Section VIII is the recommended option for abstention.

combination.

TABLE VI. PRACTITIONER DECISION FRAMEWORK

Use Case	Recommended CP Method	Rationale
Balanced, no fairness req.	Standard CP	Smallest sets, simplest
Imbalanced, fairness req.	Class-Cond CP	Per-class coverage guarantee
Custom subgroups	Mondrian CP	Group-conditional coverage
Adaptive set sizes	APS	Adaptive to difficulty
High-stakes medical	CC CP + Reject	Fairness + abstention

X. LIMITATIONS

There are a number of caveats to be noted. First, all three datasets are binary, and it is left to future for multi-class extension. work. performance is strictly limited to two class problems. Class-Conditional and Mondrian CP are theoretically applicable to multi-class problems, with the performance of the two being strictly limited to two-class problem. behaviour in the case of many classes and small calibration sample sizes per class needs a separate investigation. The trade-off between the ceiling correction in the per-class quantile and n_k is quite aggressive.

The coverage and size of the sets may change from one class to another. The cross-dataset transfer study also involved projecting onto the first four features, discarding most of the features. However, the information contained in the higher dimensional datasets (30 features for the WDBC and 14 features for the EEG). This is the reason for the low spot Transfer experiments' accuracy and restricts conclusions on CP's robustness to distribution. shift. A more principled approach would involve using domain adaptation or representation learning to map features between.

Avoid using datasets prior to applying CP. Third, the exchangeability assumption might fail when the medical data is time-ordered: patient characteristics and disease outcomes may change over time within a patient, potentially violating the exchangeability assumption. The frequency of diseases change with time. The online conformal prediction methods [18] address this by updating the calibration. Data sets were not assessed here but will be set as new data comes in. Fourth, the same random seed (42) is used for all experiments (Author garbles).

The relative Absolute metric values might slightly differ with seeds but are expected to be stable across seeds; a multi-seed analysis with confidence intervals would make statistical claims more solid. Last but not least, I did not have a chance to review newer CP variants like Regularised APS (RAPS) [19] or Kandinsky CP These foundation methods can

be improved with set-size-coverage trade-offs similar to that of framework [8] that may be better. A comparison with these newer methods is a natural next step.

XI. CONCLUSION

In this paper, we have systematically compared, empirically, four baseline conformal prediction families: Standard, Class-Conditional, Mondrian, Adaptive Prediction Sets – on three moderately imbalanced tabular medical problems. Five different base classifiers and five different coverage levels were used for datasets. The 300 main configurations, complemented by A cross-dataset transfer study and reject-option study yield five take-away findings. First, the finite-sample coverage guarantee is true in practice: 91.7 percent of all configurations have empirical coverage coverage that is equal to or greater than the target.

Second, the Standard CP gives the most efficient prediction sets (mean size), But shows the greatest class conditional coverage gap (0.103) but is less well suited for fairness-sensitive applications. Third, Class-Conditional and Mondrian CP have the best fairness-coverage trade-off. They are the default option for moderately unbalanced medical set if no other is suggested by the user, and they are recommended for default use for moderately imbalanced medical sets (gap of 0.022 with set size of 1.401).

classification. Fourth, APS is overly conservative (99.87 percent empirical coverage at the 95 percent target) and produces Competitive efficiency is best achieved in combination with regularised variants (e.g., RAPS) because unnecessary large prediction sets. Fifth, reject improves accepted-prediction accuracy by as much as 14.46 percentage points on the Blood Transfusion To transform CP into a comprehensive decision support system, this dataset is required. This dataset is necessary to make CP a full decision support system.

The limiting factor is ize. The second is comparison to newer CP variants like RAPS and Kandinsky. Not evaluated here were the CP, which are the most important items. The third one is online CP under concept drift, in which the exchangeability assumption is Time-ordered data violated by. This paper’s from scratch implementation serves as a basis for these. extensions, and that it will be a resource for both research and teaching for the increasingly large number of practitioners. Conformal prediction for safety critical applications.

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